Mental-health emergency visits, community follow-up and acute psychiatric beds across Chile's Health Services, 2024

C1 · Business Intelligence (IIB423T-1) · Universidad del Desarrollo

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Problem and decision

Main question. In 2024, how did the 29 Health Services differ in mental-health emergency visits, community follow-up and acute psychiatric bed use, and which area deserves a closer look in each?

Proposed user. Mental-health unit of a Health Service; MINSAL network division.

Decision. Review first: (a) community follow-up, (b) acute bed capacity, (c) organisation of emergency and referral care.

Scope. Public network, 2024, visits and bed-days, **not people**.

From W1 to C1

W1	C1, after feedback
Stress ratio: visits / bed-days	Visits, bed-days and occupancy shown separately
Law 21.331 as a mandate being breached	Law and ENS 2030 as context only
"15% target", "2.4% budget"	Removed: not in the sources
Emergency vs beds	+ community follow-up (REM A06)
640 vs 609 facilities	Resolved: 609 legacy + 31 modern codes

Data: three sources, one territorial code

Pathway stage	Source	One row =	Observed in 2024
Community care	REM A06 section A.1	facility × month × professional	Yes: 5.3 M follow-up visits
Emergency visit	SADU causes 36, 35	facility × day × cause	Yes: 305,431 + 9,753 visits
Decision to hospitalise	SADU cause 42	facility × day	Yes: 27,306, hospital EDs only
Waiting for a bed	SADU causes 27, 28	facility × day	**No: zero on every facility-day**
Acute psychiatric stay	REM20, 4 acute areas	hospital × area × month	Yes: 1,130 beds, 89.3% occupied

No patient identifier: sources are aggregated to Health Service × 2024 and compared side by side, not joined row by row.

Facility join SADU → registry: 640/640 facilities, N:1, no duplication. Same 29 Health Service codes in all sources; names differ (e.g. "Iquique" = "Tarapacá").

Quality checks that changed the analysis

- **Cause hierarchy.** Cause 36 = 37+38+39+40+41 on all 224,330 facility-days. 36 is the mental-health total and includes suicidal ideation. W1's 35+37+39+40 mixed levels.
- **Cause 42** is not a visit: 25 = 7+8+22+23+32+33+42 on every facility-day, so 42 = ED hospitalisations for mental disorders.
- **Recorded zeros.** "Patients waiting for a bed" is zero everywhere, so waiting cannot be measured.
- **Acute beds.** Four acute areas (mean stay 17–26 days); long-stay and forensic areas excluded. W1 had missed psychiatric intensive care.
- **Cleaning.** 24 duplicate rows removed, 28 codes stripped, 1 facility's blank Health Service filled from the registry. Occupancy recomputed from bed-days matches the published index.

National picture, 2024

1.57%

of 19.4 M emergency visits are mental-health visits

17.3

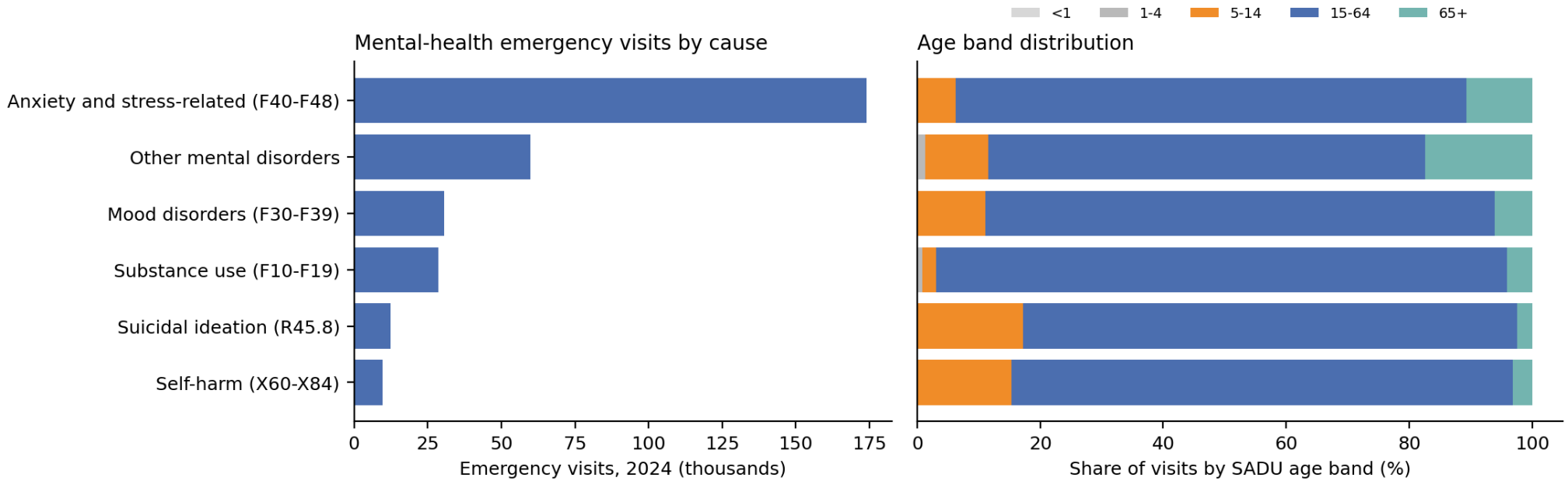
ED hospitalisations per 100 mental-health visits at hospital EDs

89.3%

acute psychiatric occupancy (1,130 beds on average)

17.4

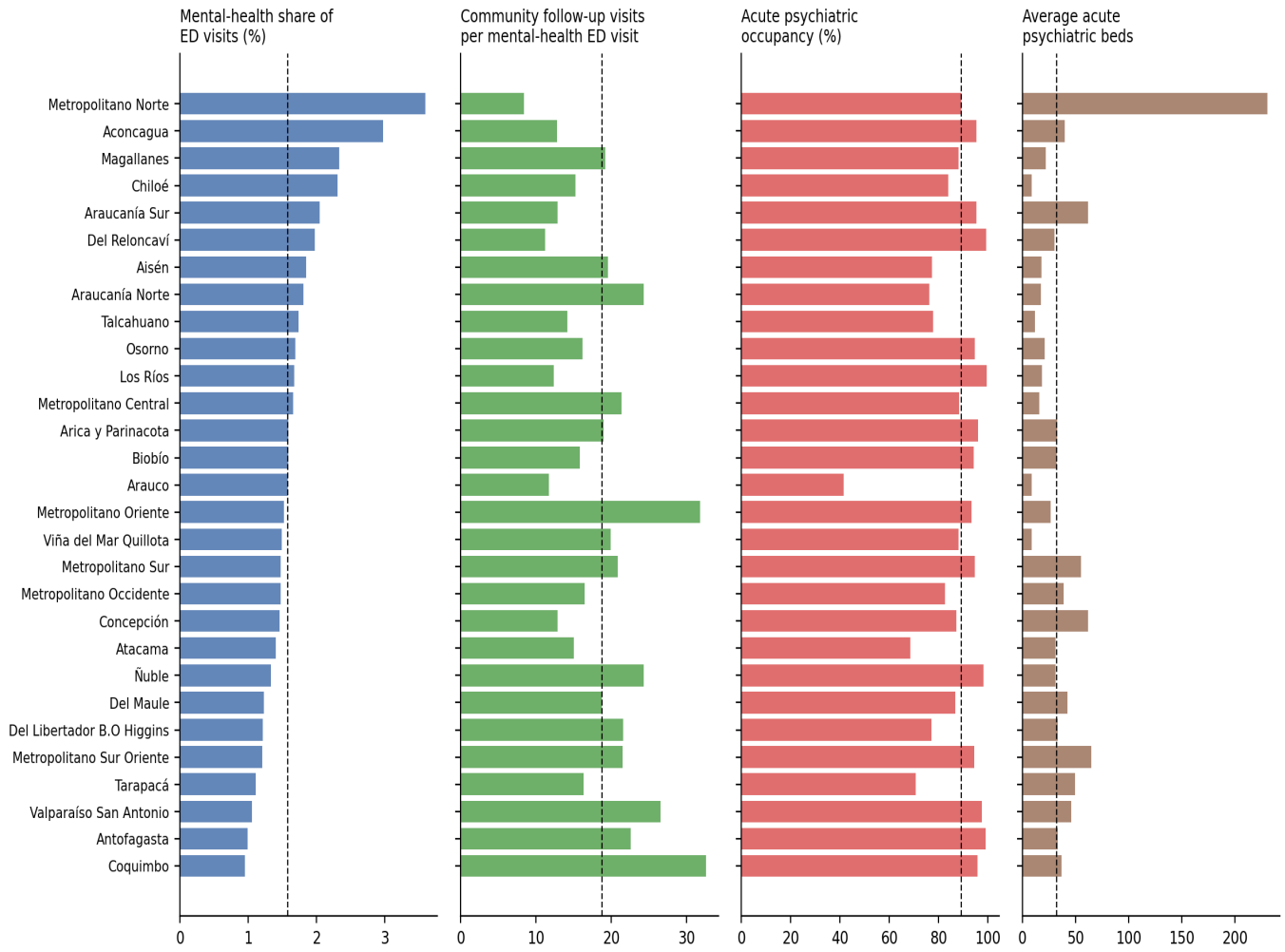
community follow-up visits per mental-health ED visit



Source: DEIS Atenciones de Urgencia 2024. Visits, not people. 15-64 is the narrowest adult band SADU publishes.

Anxiety/stress-related: 57% of visits. 5–14 band: 17% of suicidal ideation, 15% of self-harm. 15–64 includes ages 15–17.

Health Service profiles



Dashed line: median of the 29 Health Services. Sources: DEIS SADU 2024, REM20 2024 (acute psychiatric areas), REM A06 section A.1 2024.

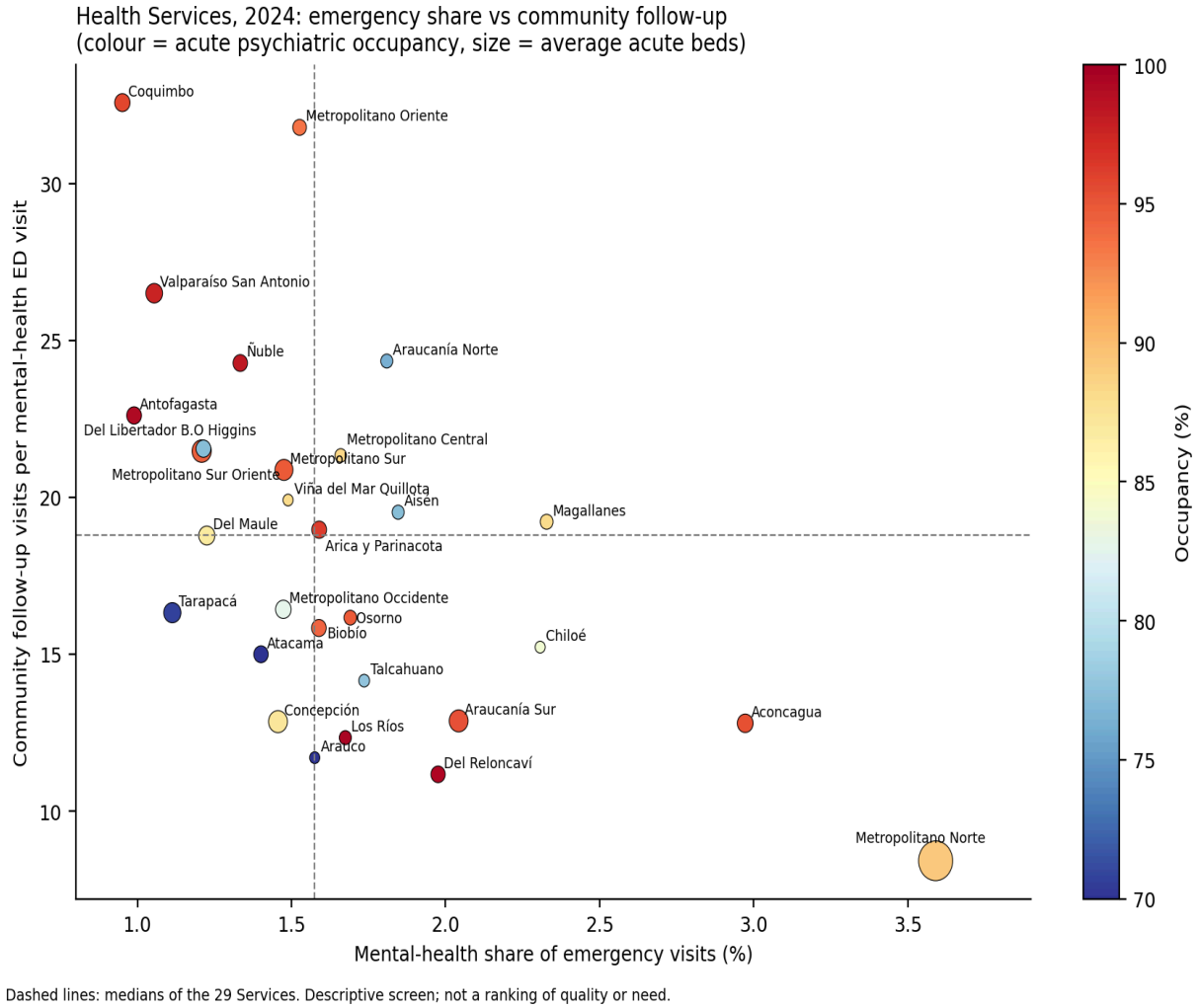
Follow-up per ED visit: 8.4 to 32.6 (median 18.8).

Occupancy: 41.5% to 99.6%.

Concentration. The Horwitz Institute has 55% of Metropolitano Norte's mental-health visits; Philippe Pinel has 61% of Aconcagua's. Without them the shares fall from 3.6% to 1.7% and from 3.0% to 1.2%, so the high shares reflect referral centres, not local need.

Check before reading as care: Araucanía Norte 0.7 ED hospitalisations per 100 (median 21.1); Arauco 41.5% occupancy.

Screen: where to look first



Flags: emergency share above the median (without psychiatric EDs), follow-up below the median, occupancy ≥ 90%.

Flags	Services
3	Biobío, Araucanía Sur, Los Ríos, Osorno, Del Reloncaví
2	Arica y Parinacota, Aconcagua, Metropolitano Norte, Talcahuano, Arauco, Chiloé
0	Viña del Mar Quillota, O'Higgins, Del Maule

Occupancy flag: 20 Services at 85%, 14 at 90%, 9 at 95%. A descriptive screen, not a ranking of quality or need.

What the evidence supports, and what it does not

Observed. Several-fold differences in follow-up and occupancy. Five Services combine a high emergency share, low follow-up and full beds.

Interpretation. For those five, review community follow-up first, then bed management.

Hypotheses. Low follow-up leads to more crises reaching the ED; long stays keep occupancy high.

Alternatives. Recording practice; referrals between Services; private sector; population size and age.

Cannot establish: need, quality, waiting time, causality, or whether a person was followed up after an ED visit.

W1 reconciled. Del Maule 1.83 reproduces, but is 1.17 with acute beds. "18 Services without child beds" is reversed: 18 have child short-stay beds; 7 report no acute child area (not reported ≠ no service).

Needed next: linked ED-admission records, a population denominator, 2021–2023 data, confirmation of anomalies.

Continuity: machine learning and dashboard

ML: clustering Health Service profiles

- Unit: Health Service × year, 2021–2024 (116 rows; data already available).
- Inputs: the five indicators, primary-care share of follow-up, child share of acute beds.
- No labels; clusters described. A Service compares itself with similar peers.
- For review: an occupancy ≥ 95% month, predicted from previous months only (no same-month leakage).

Dashboard

- Users: Health Service mental-health unit; MINSAL network division.
- Views: screen scatter; Service profile by month; data-quality panel.
- Filters: year, month, cause, age band, facility type, with or without psychiatric EDs.
- Waiting left out until a source records it.

Process record

Feedback response. 9 of 10 items accepted, 1 adapted: the DEIS 2024 establishment list could not be downloaded from our environment, so the 2026 registry was used and the gap stated.

AI use. W1: Gemini (scaffolding, translation). C1: Claude (extraction, notebook, drafting). Unsupported W1 claims removed after source checks. Every number comes from the executed notebook, and Service totals reconcile with national totals.

Contributions

Member	Contribution
Benjamín Buzeta	W1 context and report. C1: feedback response, source checks, report
Oscar Pichuman Millape	W1 extraction, notebook and joins. C1: review of extraction, profiling and bed definition
Felipe Alfredo Tajan Gebauer	W1 feedback questions. C1: REM A06 indicator, Service profiles and screen, ML/dashboard, slides